Essentials of
economics

Krugman • Wells • Olney

chapter 2

Economic Models: Trade-offs and Trade

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chapter 2

TUNNEL VISION



Clearly, the Wright brothers believed in their model.

What you will learn in this chapter:

- Why **models**—simplified representations of reality—play a crucial role in economics
- Three simple but important models: the **production possibility frontier, comparative advantage,** and the **circular-flow diagram**
- The difference between **positive economics**, which tries to describe the economy and predict its behavior, and **normative economics**, which tries to prescribe economic policy
- When economists agree and why they sometimes disagree

Models in Economics: Some Important Examples

chapter 2

A **model** is a simplified representation of a real situation that is used to better understand real-life situations.

The **other things equal assumption** means that all other relevant factors remain unchanged.

Ceteris Paribus



Alfred Marshall

Models in Economics: Some Important Examples

chapter 2

Trade-offs: The Production Possibility Frontier

The **production possibility frontier** illustrates the trade-offs facing an economy that produces only two goods. It shows the maximum quantity of one good that can be produced for any given quantity produced of the other.



What to do? Even a castaway faces trade-offs.

Models in Economics: Some Important Examples

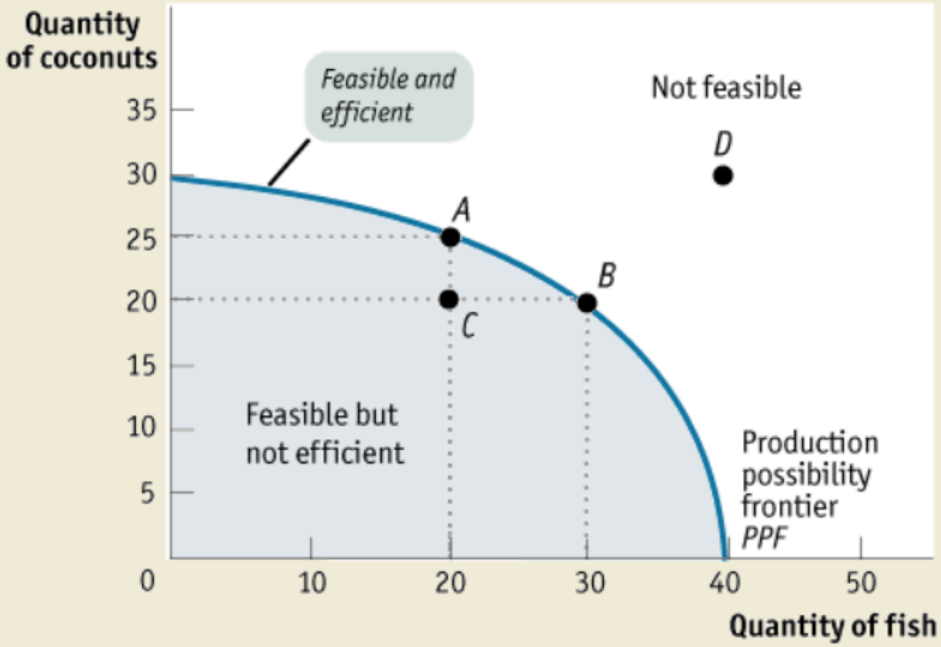
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Trade-offs: The Production Possibility Frontier

Figure 2-1

The Production Possibility Frontier

The production possibility frontier illustrates the trade-offs facing an economy that produces two goods. It shows the maximum quantity of one good that can be produced given the quantity of the other good produced. Here, the maximum number of coconuts that Tom can gather depends on the number of fish he catches, and vice versa. His feasible production is shown by the area *inside or on* the curve. Production at point *C* is feasible but not efficient. Points *A* and *B* are feasible and efficient, but point *D* is not feasible.



Models in Economics: Some Important Examples

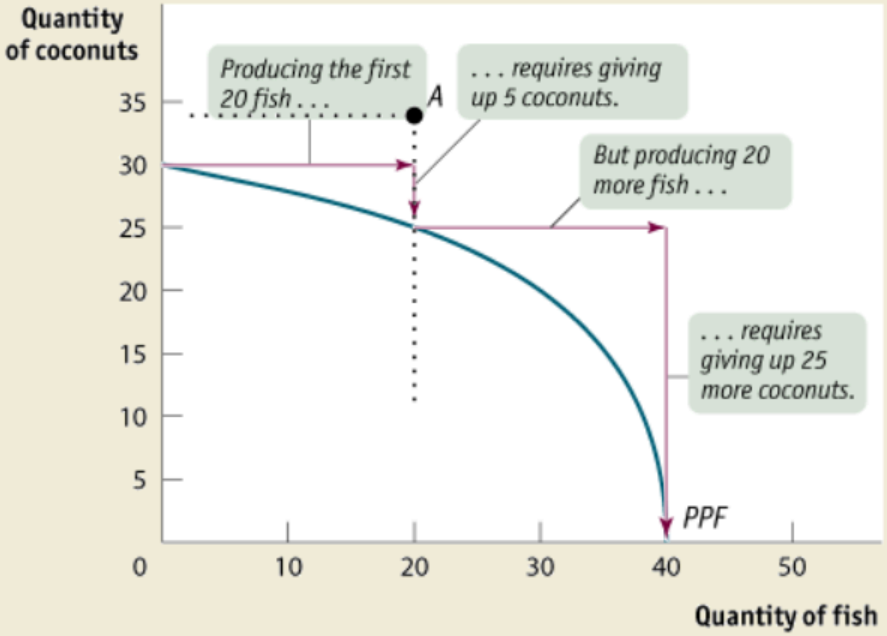
chapter 2

A PPF shows two things: (1) Trade-offs (2) Opportunity cost

Figure 2-2

Increasing Opportunity Cost

The bowed-out shape of the production possibility frontier reflects increasing opportunity cost. In this example, to produce the first 20 fish, Tom must give up 5 coconuts. But to produce an additional 20 fish, he must give up 25 more coconuts.



Models in Economics: Some Important Examples

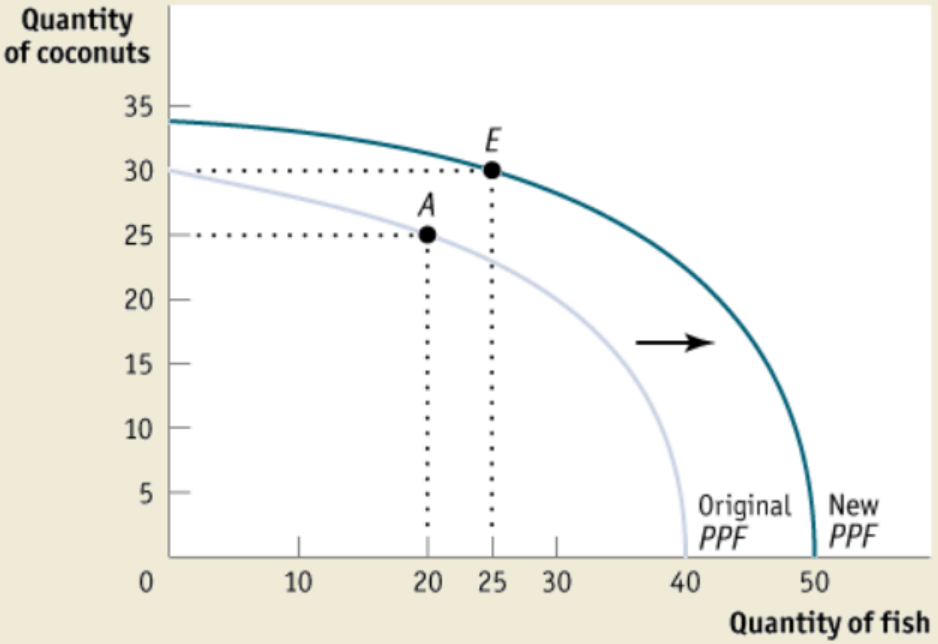
chapter 2

Trade-offs: The Production Possibility Frontier

Figure 2-3

Economic Growth

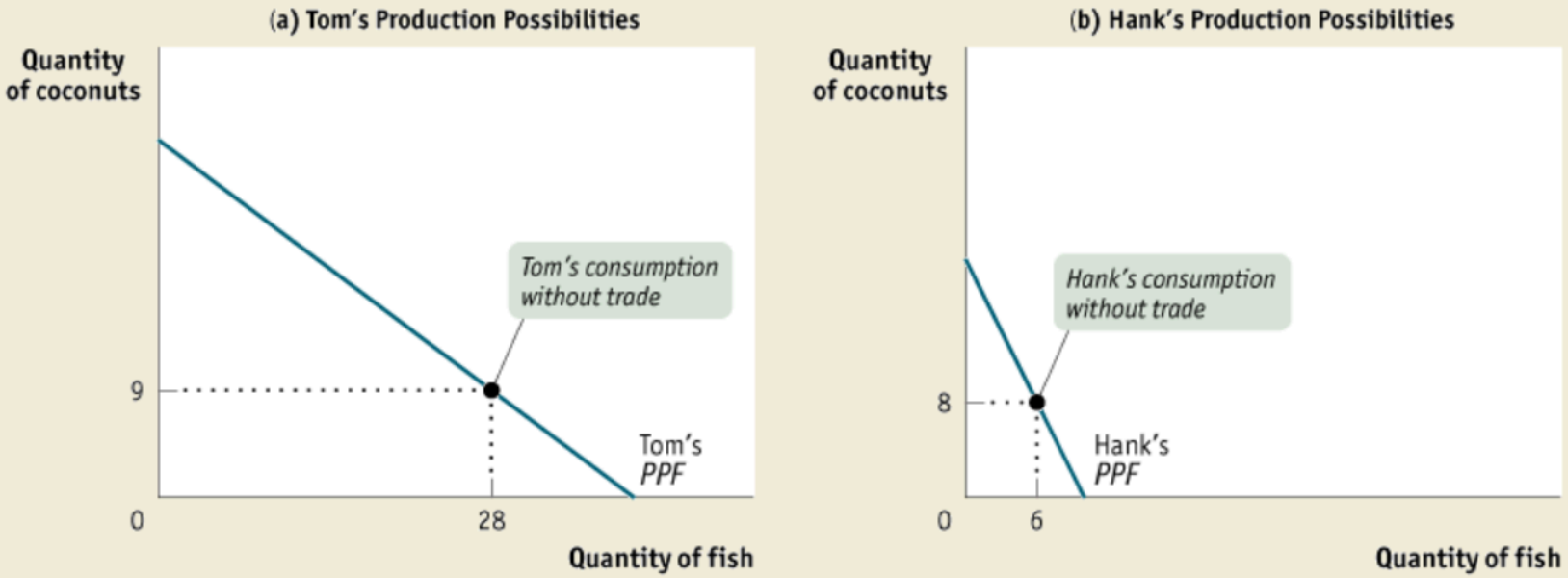
Economic growth results in an *outward shift* of the production possibility frontier because production possibilities are expanded. The economy can now produce more of everything. For example, if production is initially at point *A* (20 fish and 25 coconuts), it could move to point *E* (25 fish and 30 coconuts).



Models in Economics: Some Important Examples

Comparative Advantage and Gains from Trade

Figure 2-4 Production Possibilities for Two Castaways



Here, each of the two castaways has a constant opportunity cost of fish and a straight-line production possibility frontier: In Tom's case, each fish always has an opportunity cost of 3/4 of a coconut. In Hank's case, each fish always has an opportunity cost of 2 coconuts.

Models in Economics: Some Important Examples

Comparative Advantage and Gains from Trade

TABLE **2-1**

Tom's and Hank's Opportunity Costs of Fish and Coconuts

	Tom's Opportunity Cost	Hank's Opportunity Cost
One fish	3/4 coconut	2 coconuts
One coconut	4/3 fish	1/2 fish

Tom's opportunity cost for producing fish is less than Hank's. Also, Hank's opportunity cost for producing coconuts is less than Tom's. So, Tom has comparative advantage in fish production while Hank has in coconut gathering.

Models in Economics: Some Important Examples

TABLE **2-2**

How the Castaways Gain from Trade

		Without Trade		With Trade		Gains from Trade
		Production	Consumption	Production	Consumption	
Tom	Fish	28	28	40	30	+2
	Coconuts	9	9	0	10	+1
Hank	Fish	6	6	0	10	+4
	Coconuts	8	8	20	10	+2

Is this deal possible? Tom gives Hank 10 fish and Hank gives Tom 10 coconuts (1 : 1 trade)?

Tom, by giving 10 fish, he would've been happy if he could have 7.5 (=10*(3/4)) coconuts.

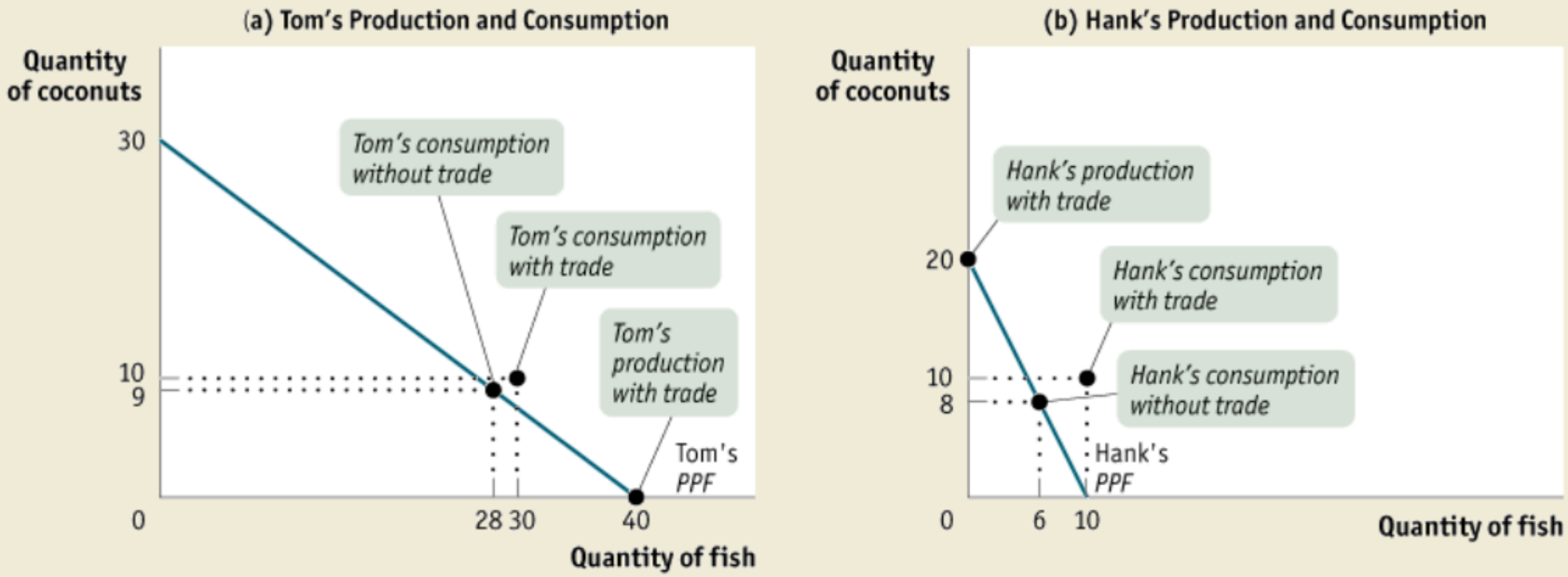
Hank would've been happy if he could have 5 (= 10*(1/2)) fish.

fish : coconuts trade ratio between 1: (3/4 ~ 2) is all possible

Models in Economics: Some Important Examples

Comparative Advantage and Gains from Trade

Figure 2-5 Comparative Advantage and Gains From Trade



By specializing and trading, the two castaways can produce and consume more of both goods. Tom specializes in catching fish, his comparative advantage, and Hank—who has an *absolute* disadvantage in both goods but a *comparative* advantage in coconuts—specializes in gathering coconuts. The result is that each castaway can consume more of both goods than either could without trade.

Models in Economics: Some Important Examples

Comparative Advantage and Gains from Trade

An individual has a **comparative advantage** in producing a good or service if the opportunity cost of producing the good is lower for that individual than for other people.

An individual has an **absolute advantage** in an activity if he or she can do it better than other people. Having an absolute advantage is not the same thing as having a comparative advantage.

The Full Analysis -1

Step 1: Calculate the opportunity costs

Tom's opp. cost for catching one fish is (3/4) coconuts (40 fish for 30 coconuts).
Tom's opp. cost for gathering one coconut is (4/3) fish (30 coconuts for 40 fish).
Hank's opp. cost for catching one fish is (2) fish (10 fish for 20 coconuts).
Hank's opp. cost for gathering one coconut is (1/2) fish (20 coconuts for 10 fish).

	Opportunity cost table			
	Tom		Hank	
Fish	3/4		2	coconuts
Coconuts	4/3		1/2	fish

Step 2: Find out who has comparative advantage in what.

So Tom has comparative advantage in (catching fish : 3/4 is less than 4/3) and Hank has comparative advantage in (gathering coconuts: 1/2 is less than 2).

The Full Analysis -2

Step 3: Find out the deal is possible

Lets suppose Tom catches 40 fish and give 10 to Hank for 10 coconuts, and Hank gathers 20 coconuts and give 10 to Tom for 10 fishes.

(1) Is this possible trade?

Tom gave 10 fishes to Hank and received 10 coconuts. Tom would have been happy if he could receive only (7.5 = 10 * 3/4) coconuts. So Tom is more than happy.

Hank gave 10 coconuts to Tom and received 10 fishes. Hank would have been happy if he could receive only (5 = 10 * 1/2) fishes. So Hank is also more than happy.

(2) Does this trade make both of them batter off?

Yes. They could not reach this without the trade.

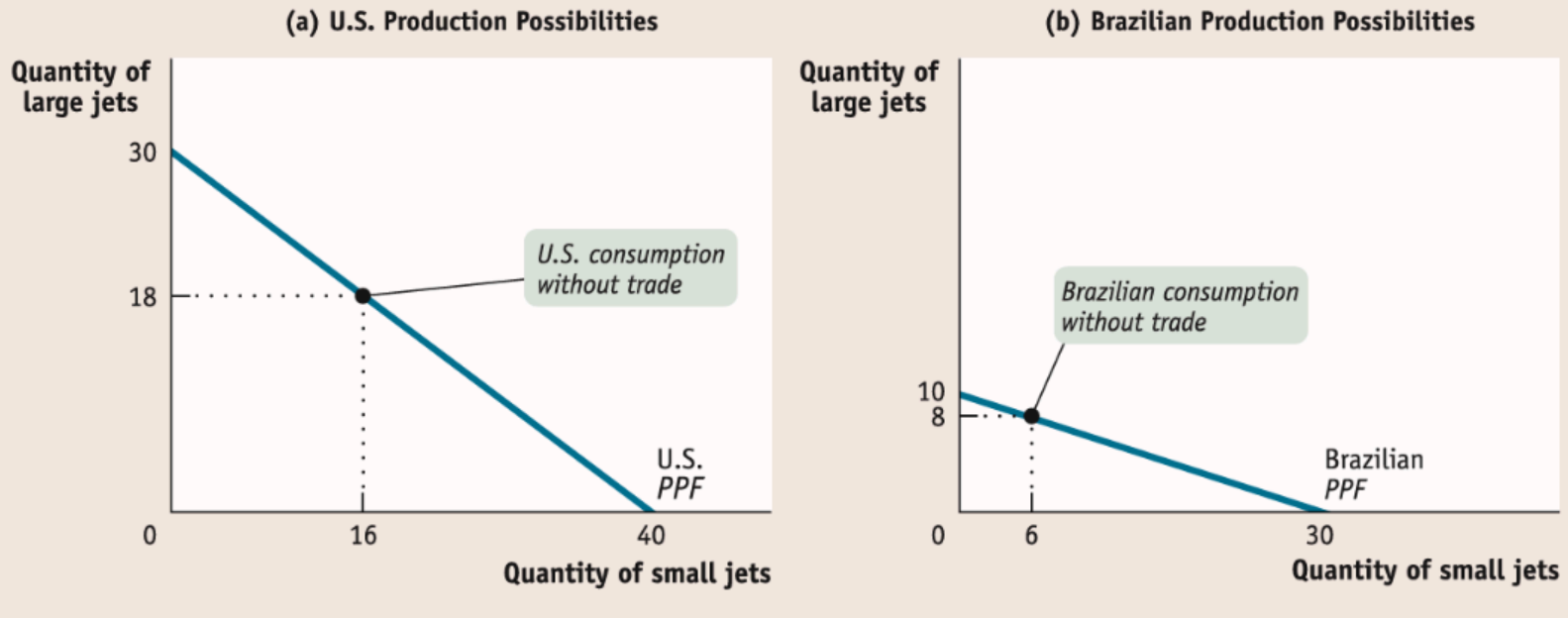
Models in Economics: Some Important Examples

Comparative Advantage

TABLE 2-1 U.S. and Brazilian Opportunity Costs of Small Jets and Large Jets

	U.S. Opportunity Cost		Brazilian Opportunity Cost
One small jet	¾ large jet	>	⅓ large jet
One large jet	4/3 small jets	<	3 small jets

FIGURE 2-4 Production Possibilities for Two Countries



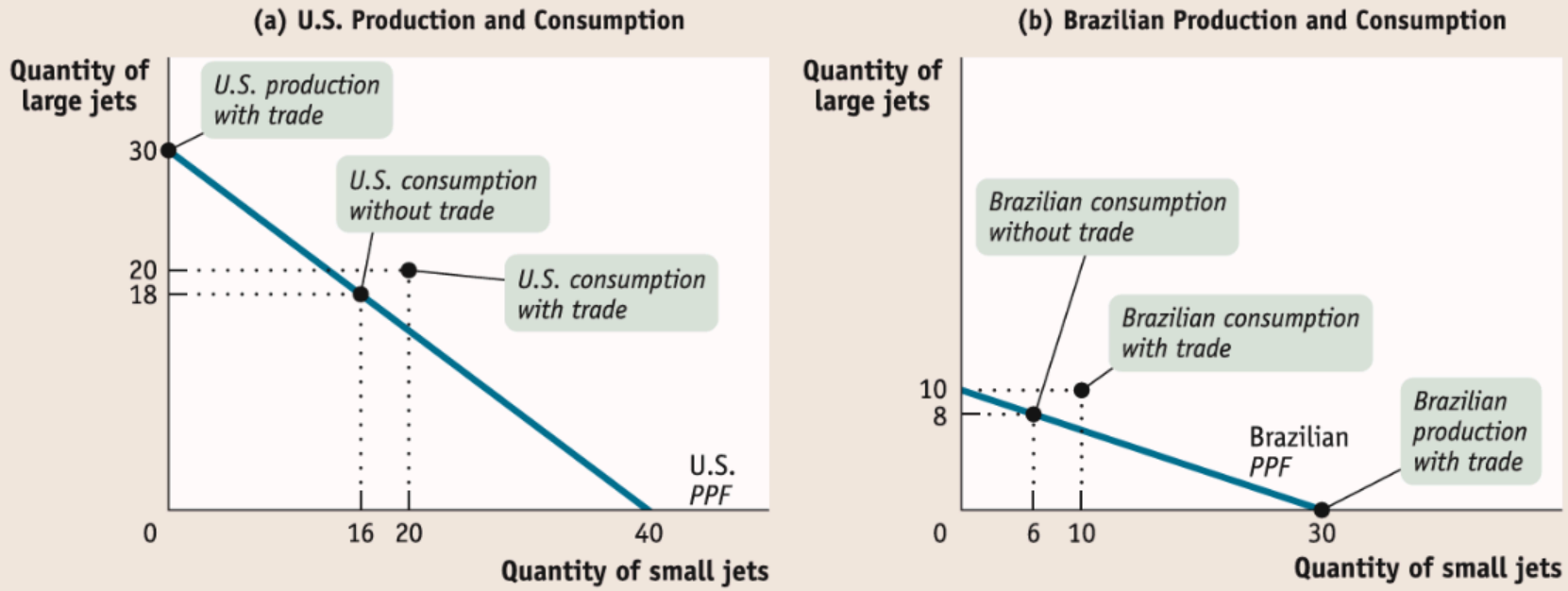
Here, both the United States and Brazil have a constant opportunity cost of small jets, illustrated by a straight-line production possibility frontier. For the United States, each small jet has an opportunity cost of ¾ of a large jet. Brazil has an opportunity cost of a small jet equal to ⅓ of a large jet.

Models in Economics: Some Important Examples

2 apter

Comparative Advantage and International Trade

FIGURE 2-5 Comparative Advantage and Gains from Trade



By specializing and trading, the United States and Brazil can produce and consume more of both large jets and small jets. The United States specializes in manufacturing large jets, its comparative advantage, and Brazil—which has an *absolute* disadvantage in both goods but a *comparative* advantage in small jets—specializes in manufacturing small jets. With trade, both countries can consume more of both goods than either could without trade.

Models in Economics: Some Important Examples

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Comparative Advantage and International Trade

TABLE 2-2 How the United States and Brazil Gain from Trade

		Without Trade		With Trade		Gains from Trade
		Production	Consumption	Production	Consumption	
United States	Large jets	18	18	30	20	+2
	Small jets	16	16	0	20	+4
Brazil	Large jets	8	8	0	10	+2
	Small jets	6	6	30	10	+4

Models in Economics: Some Important Examples

Transactions: The Circular-Flow Diagram

Trade takes the form of **barter** when people directly exchange goods or services that they have for goods or services that they want.

The **circular-flow diagram** is a model that represents the transactions in an economy by flows around a circle.

Models in Economics: Some Important Examples

Transactions: The Circular-Flow Diagram

A **household** is a person or a group of people that share their income.

A **firm** is an organization that produces goods and services for sale.

Firms sell goods and services that they produce to households in **markets for goods and services**.

Firms buy the resources they need to **produce—factors of production**—in **factor markets**.

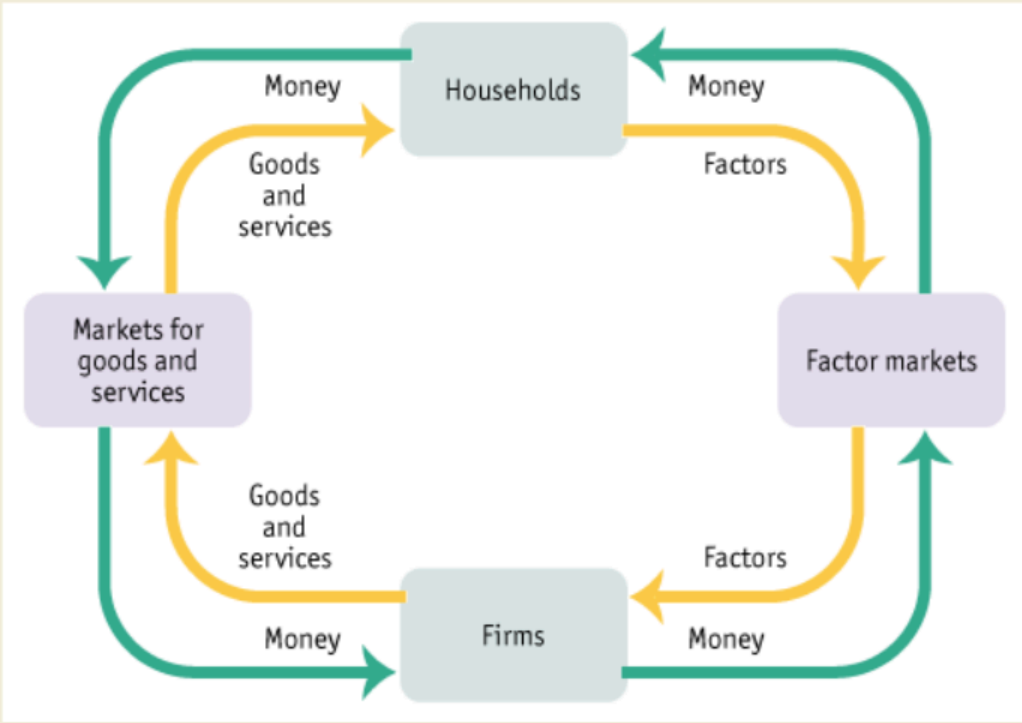
Models in Economics: Some Important Examples

Transactions: The Circular-Flow Diagram

Figure 2-7

The Circular-Flow Diagram

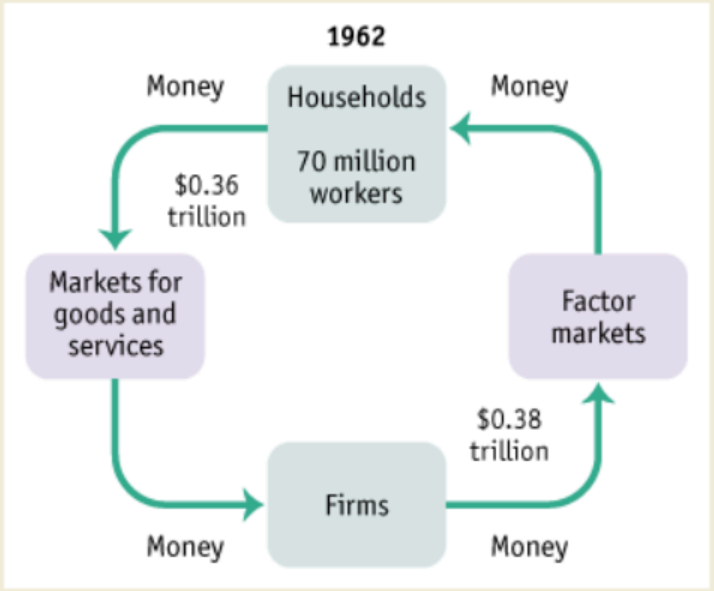
This model represents the flows of money and goods and services in the economy. In the markets for goods and services, households purchase goods and services from firms, generating a flow of money to the firms and a flow of goods and services to the households. The money flows back to households as firms purchase factors of production from the households in factor markets.

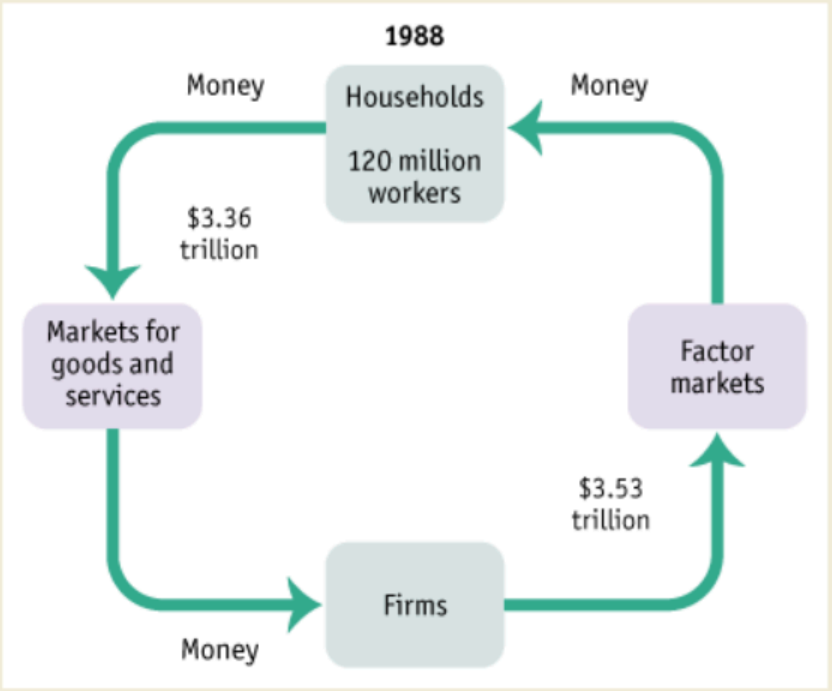


Models in Economics: Some Important Examples

Transactions: The Circular-Flow Diagram

Figure 2-8 Growth in the U.S. Economy from 1962 to 1988





These two circular-flow diagrams—one corresponding to 1962, the other corresponding to 1988—help us understand how the U.S. economy was able to produce enough jobs for its rapidly growing labor force. A roughly twofold increase in the number of workers from 1962 to 1988 was accompanied by a ninefold increase in money flows between households and firms. As the labor force grew, money going to households increased and their spending on goods and services increased. This led firms to hire more workers to meet the increased desire for goods and services and generated more jobs for households.

Using Models

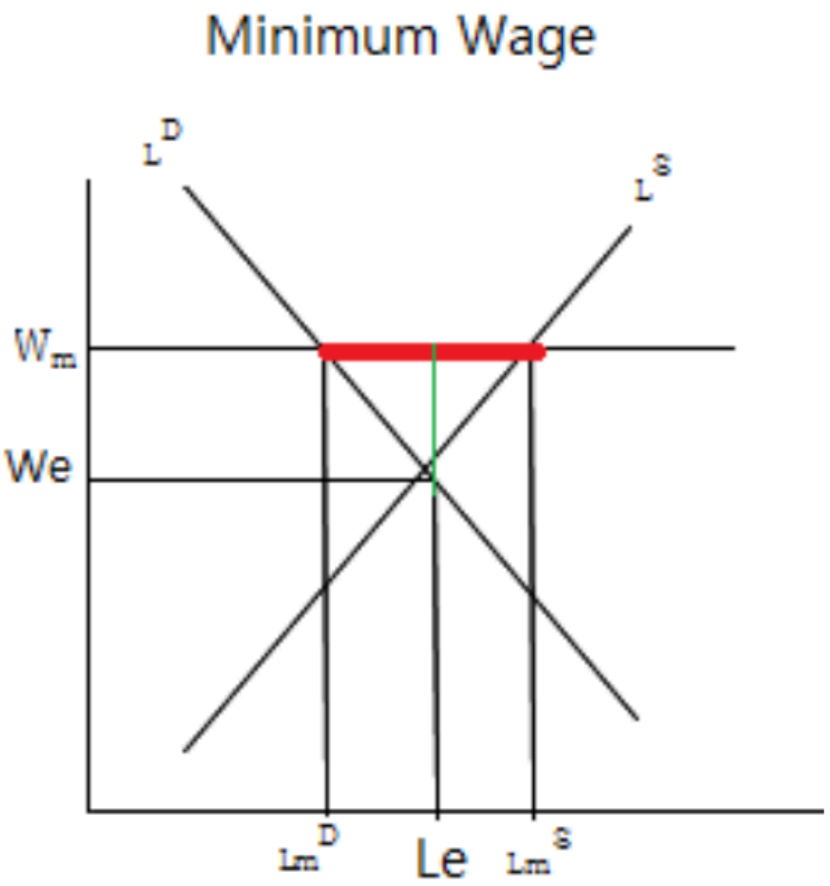
Positive versus Normative Economics

Positive economics is the branch of economic analysis that describes the way the economy actually works.

Normative economics makes prescriptions about the way the economy should work.

A **forecast** is a simple prediction of the future.

Positive Economics and Normative Economics: Minimum Wage and Its Effects



(Example) Gain from International Trade

Step 1: Opportunity cost table

	US Opportunity Cost	Brazil Opportunity Cost
One Sugar	1/4 parts	1/10 parts
One Part	4 sugars	10 sugars

Step 2: US has comparative advantage in producing parts, and Brazil has in producing sugar.

Step 3: US consumes 16 sugar and 1 part. Brazil consumes 24 sugar (= 40 – 16) and 2 parts.

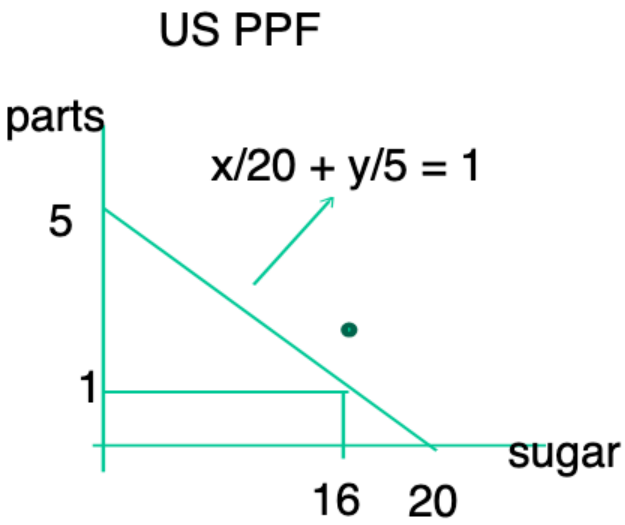
(1) Is this trade possible? US gives 2 parts to Brazil in return receives 16 sugar, and Brazil gives 16sugar and receives 2 parts

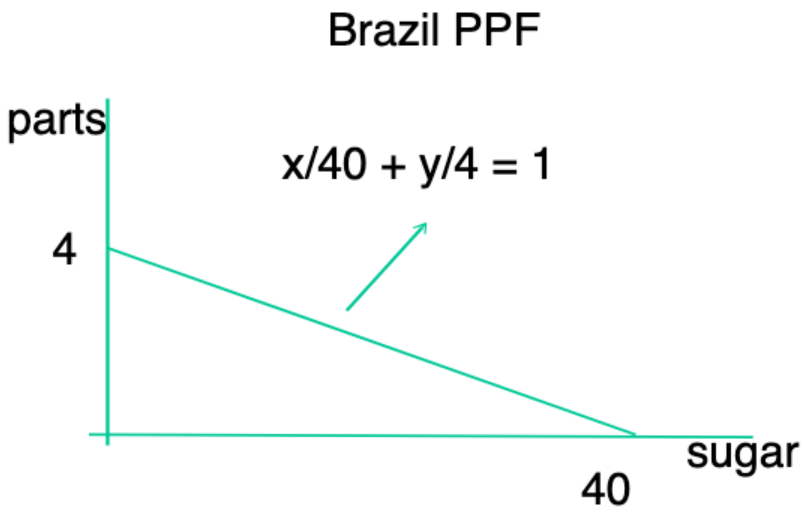
US can give up 2 parts if it can receive at least 2 * (20/4) = 10 sugar. But received 16 sugar

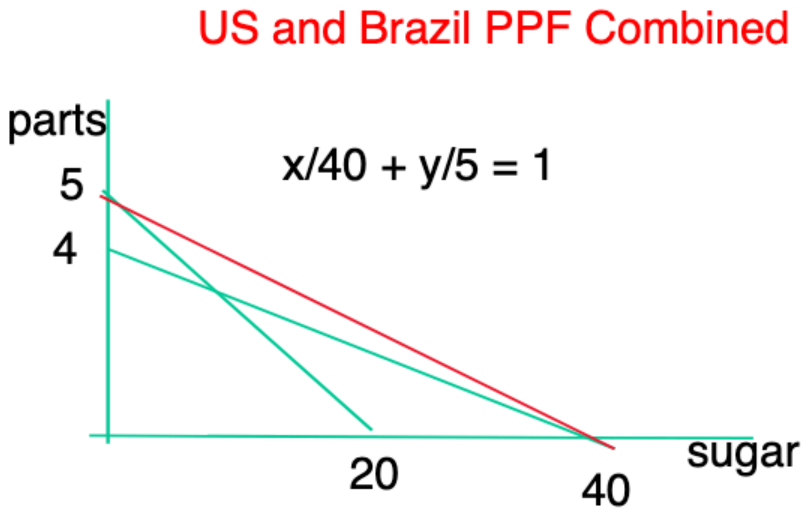
Brazil can give up 16 sugar if it receive at least 16 * (4/40) = 1.6 parts. But Brazil received 2 parts. So this trade is possible.

(2) Both countries are better off. US could not reach (16, 3), Brazil could not reach (2, 24) before.

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KEY TERMS

Model
Other things equal assumption
Production possibility frontier
Comparative advantage
Absolute advantage
Barter
Circular-flow diagram
Household
Firm
Markets for goods and services
Factor markets
Factors of production
Positive economics
Normative economics
Forecast

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>> Chapter 2 Appendix: Graphs in Economics

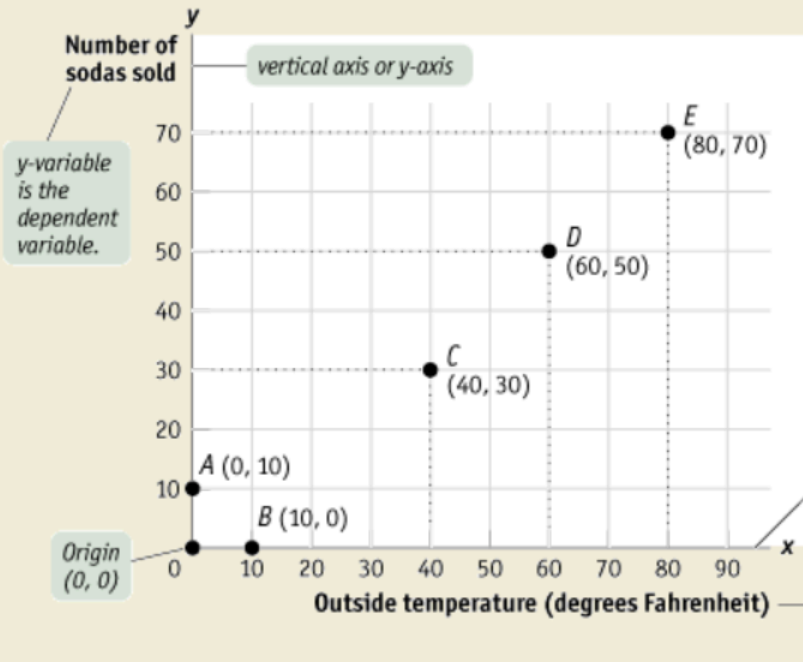
A quantity that can take on more than one value is called a **variable**.

A **causal relationship** exists between two variables when the value taken by one variable directly influences or determines the value taken by the other variable. In a causal relationship, the determining variable (usually X) is called the **independent variable**; the variable it determines is called the **dependent variable** (usually Y).

How Graphs Work

Two-Variable Graphs

Figure 2A-1 Plotting Points on a Two-Variable Graph



x-variable: outside temperature	y-variable: number of sodas sold	Point
0°F	10	A
10	0	B
40	30	C
60	50	D
80	70	E

The data from the table are plotted where outside temperature (the independent variable) is measured along the horizontal axis and number of sodas sold (the dependent variable) is measured along the vertical axis. Each of the five combinations of temperature and sodas sold is represented by a point: *A*, *B*, *C*, *D*, and *E*. Each point in the graph is identified by a pair of values. For example, point C corresponds to the pair (40, 30)—an outside temperature of 40°F (the value of the *x*-variable) and 30 sodas sold (the value of the *y*-variable).

How Graphs Work

Curves on a Graph

A **curve** is a line on a graph that depicts a relationship between two variables. It may be either a straight line or a curved line. If the curve is a straight line, the variables have a **linear relationship**. If the curve is not a straight line, the variables have a **nonlinear relationship**.

How Graphs Work

Curves on a Graph

Two variables have a **positive relationship** when an increase in the value of one variable is associated with an increase in the value of the other variable. It is illustrated by a curve that slopes upward from left to right.

Two variables have a **negative relationship** when an increase in the value of one variable is associated with a decrease in the value of the other variable. It is illustrated by a curve that slopes downward from left to right.

The **horizontal intercept** of a curve is the point at which it hits the horizontal axis; it indicates the value of the x-variable when the value of the y-variable is zero.

The **vertical intercept** of a curve is the point at which it hits the vertical axis; it shows the value of the y-variable when the value of the x-variable is zero.

A Key Concept: The Slope of a Curve

The **slope** of a line or curve is a measure of how steep it is. The slope of a line is measured by "rise over run"—the change in the *y*-variable between two points on the line divided by the change in the *x*-variable between those same two points.

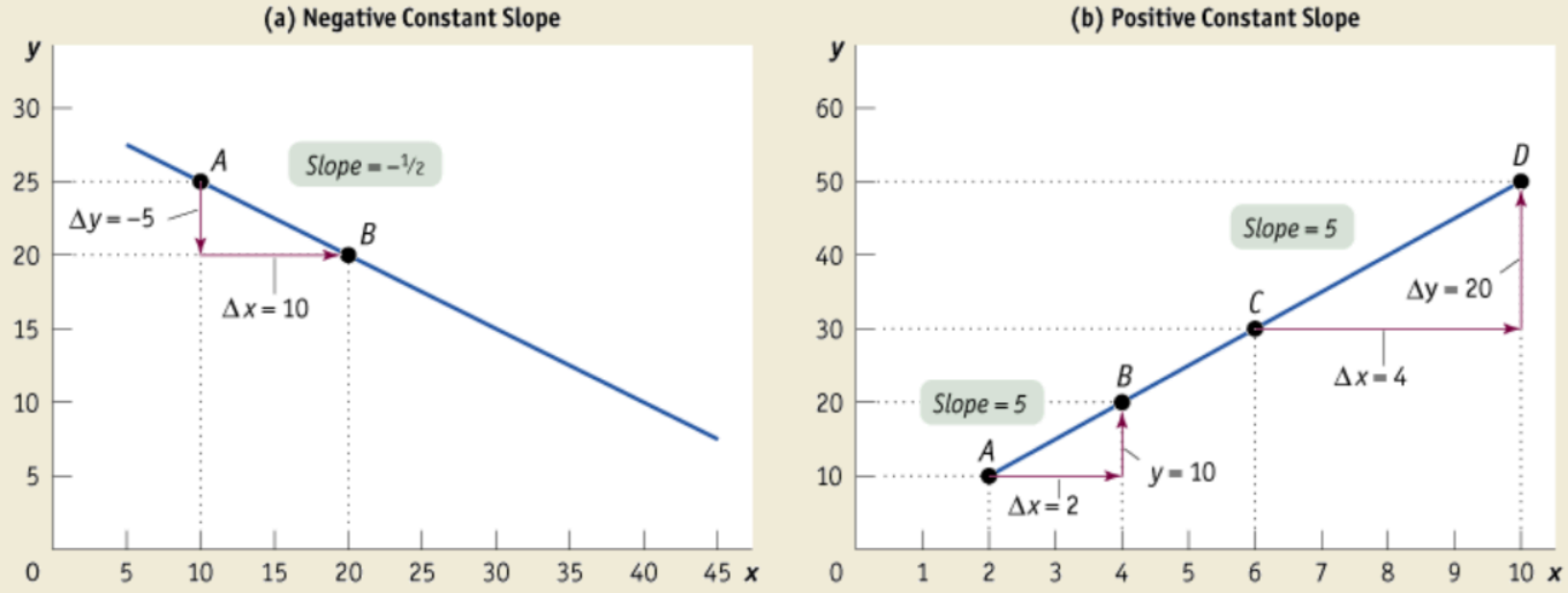
The Slope of a Linear Curve

$$\frac{\text{Change in y}}{\text{Change in x}} = \frac{\text{D}y}{\text{D}x} = \text{Slope}$$

A Key Concept: The Slope of a Curve

The Slope of a Linear Curve

Figure 2A-3 Calculating the Slope



Panels (a) and (b) show two linear curves. Between points A and *B* on the curve in panel (a), the change in *y* (the rise) is –5 and the change in *x* (the run) is 10. So the slope from *A* to *B* is $\Delta y/\Delta x = -5/10 = -1/2 = -0.5$, where the negative sign indicates that the curve is downward sloping. In panel (b), the curve has a slope from *A* to *B* of $\Delta y/\Delta x = 10/2 = 5$. The slope from *C* to *D* is $\Delta y/\Delta x = 20/4 = 5$. The slope is positive, indicating that the curve is upward sloping. Furthermore, the slope between *A* and *B* is the same as the slope between *C* and *D*, making this a linear curve. The slope of a linear curve is constant: it is the same regardless of where it is calculated along the curve.

A Key Concept: The Slope of a Curve

Horizontal and Vertical Curves and Their Slopes

When a curve is horizontal, the value of y along that curve never changes—it is constant. The slope of a horizontal curve is always zero.

If a curve is vertical, the value of x along the curve never changes—it is constant. The slope of a vertical line is equal to infinity.

A vertical or a horizontal curve has a special implication: it means that the *x*-variable and the *y*-variable are unrelated.

A Key Concept: The Slope of a Curve

The Slope of a Nonlinear Curve

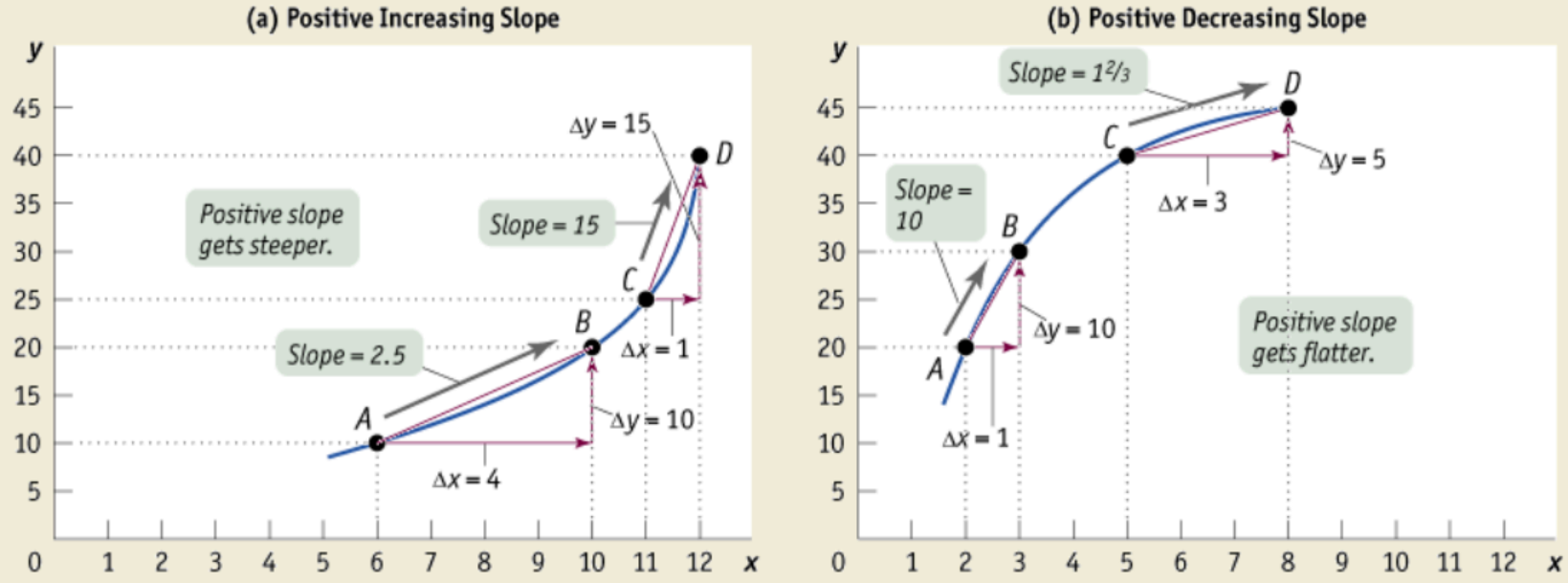
A **nonlinear curve** is one in which the slope is not the same between every pair of points.

The **absolute value** of a negative number is the value of the negative number without the minus sign.

A Key Concept: The Slope of a Curve

The Slope of a Nonlinear Curve

Figure 2A-4 Nonlinear Curves



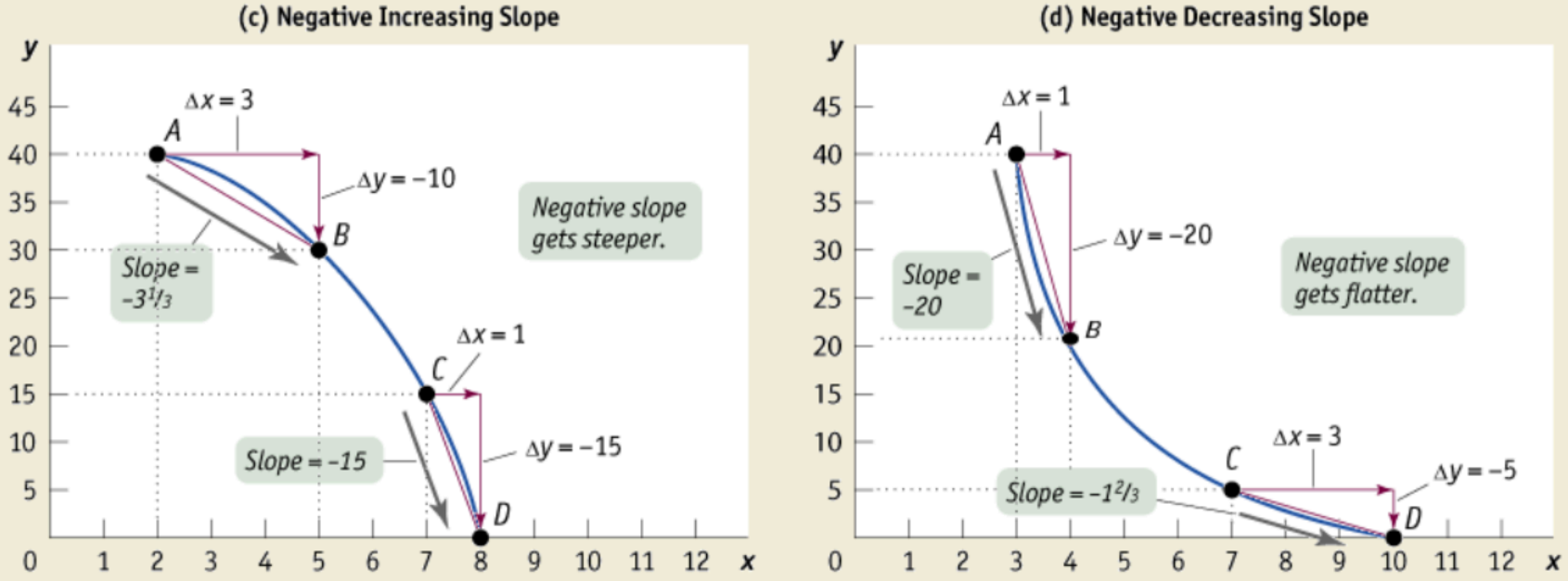
In panel (a) the slope of the curve from *A* to *B* is $\Delta y/\Delta x = 10/4 = 2.5$, and from *C* to *D* it is $\Delta y/\Delta x = 15/1 = 15$. The slope is positive and increasing; it gets steeper as you move to the right.

In panel (b) the slope of the curve slope from *A* to *B* is $\Delta y/\Delta x = 10/1 = 10$, and from *C* to *D* it is $\Delta y/\Delta x = 5/3 = 12/3$. The slope is positive and decreasing; it gets flatter as you move to the right.

A Key Concept: The Slope of a Curve

The Slope of a Nonlinear Curve

Figure 2A-4 Nonlinear Curves, cont'd.



In panel (c) the slope from *A* to *B* is $\Delta y/\Delta x = -10/3 = -3\frac{1}{3}$, and from *C* to *D* it is $\Delta y/\Delta x = -15/1 = -15$. The slope is negative and increasing; it gets steeper as you move to the right.

And in panel (d) the slope from *A* to *B* is $\Delta y/\Delta x = -20/1 = -20$, and from *C* to *D* it is $\Delta y/\Delta x = -5/3 = -12/3$. The slope is negative and decreasing; it gets flatter as you move to the right.

The slope in each case has been calculated by using the arc method—that is, by drawing a straight line connecting two points along a curve. The average slope between those two points is equal to the slope of the straight line between those two points.

First derivative of a function

When y = f(x), and its first derivative is f '(x),

f '(x) > 0 : function f(x) is increasing

f '(x) < 0 : function f(x) is decreasing

f '(x) = 0 : function f(x) is in the middle

First derivative of a function

When y = f(x), and its first derivative is f '(x),

f '(x) > 0 : function f(x) is increasing

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f '(x) = 0 : function f(x) is in the middle

First derivative of a function

= The slope of the tangent line

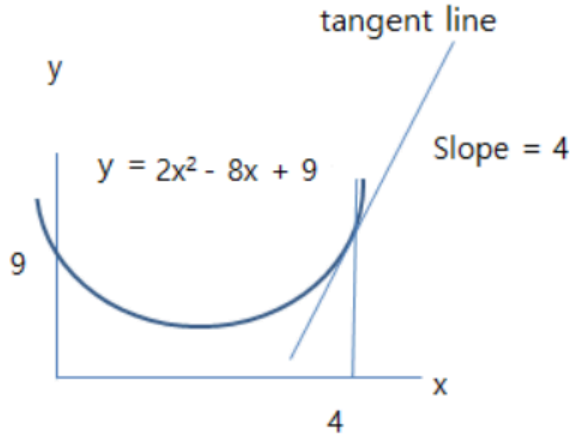
A Key Concept: The Slope of a Curve

The slope of curve = slope of the tangent line

When $y = f(x)$, then the slope of the curve given $x = x_0$ is

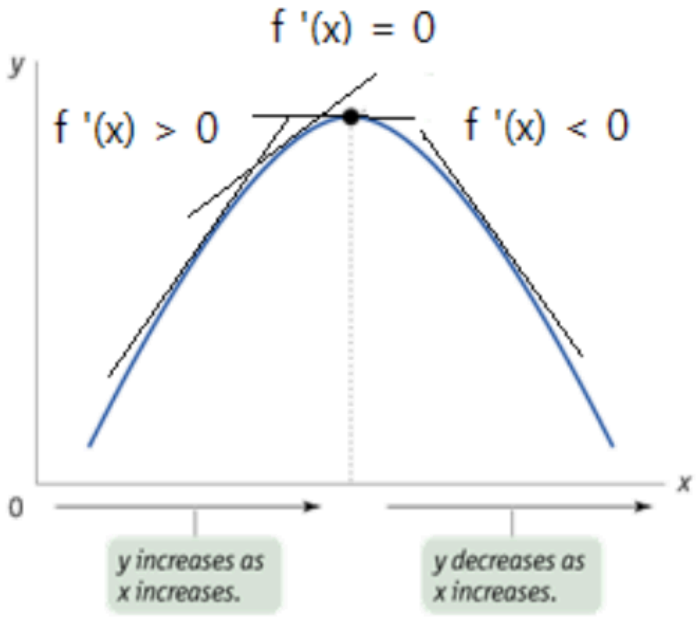
$f'(x_0)$, where $f'(x)$ is the first derivative of $f(x)$.

Example: Given $y = 2x^2 - 8x + 9$, what is the curve at $x = 3$:
Answer: $y' = f'(x) = 4x - 8$, so when $x = 3$, y' is 4.
This is the slope of the tangent line at $x = 3$



Maximum condition

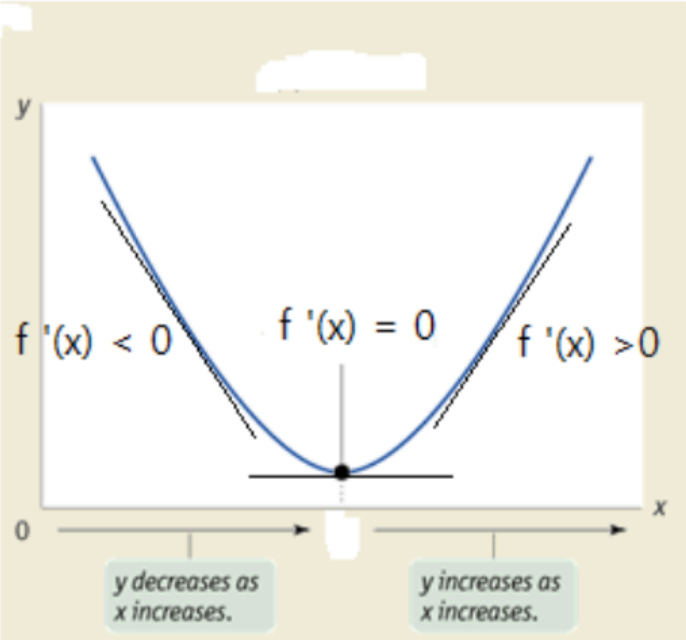
f ‘ (x) = 0, and f ‘’(x) < 0



Since f ‘(x) is changing from positive to zero to then negative, it is decreasing. So the sign of the first derivative of the first derivative function (the second derivative function from the original function f(x)) must be negative: f ‘’ (x) < 0

Minimum condition

$f'(x) = 0$, and $f''(x) > 0$



Since $f'(x)$ is changing from negative to zero to then positive, it is increasing. So the sign of the first derivative of the first derivative function (the second derivative function from the original function f(x)) must be positive: $f''(x) > 0$

Mathematics used in Economics

Please don't be horrified! This is just to show you how simple mathematics is used in Economics.

Link to the mathematics in Economics

Presenting Numerical Information

Types of Numerical Data: Time-Series Data and Cross Sectional Data

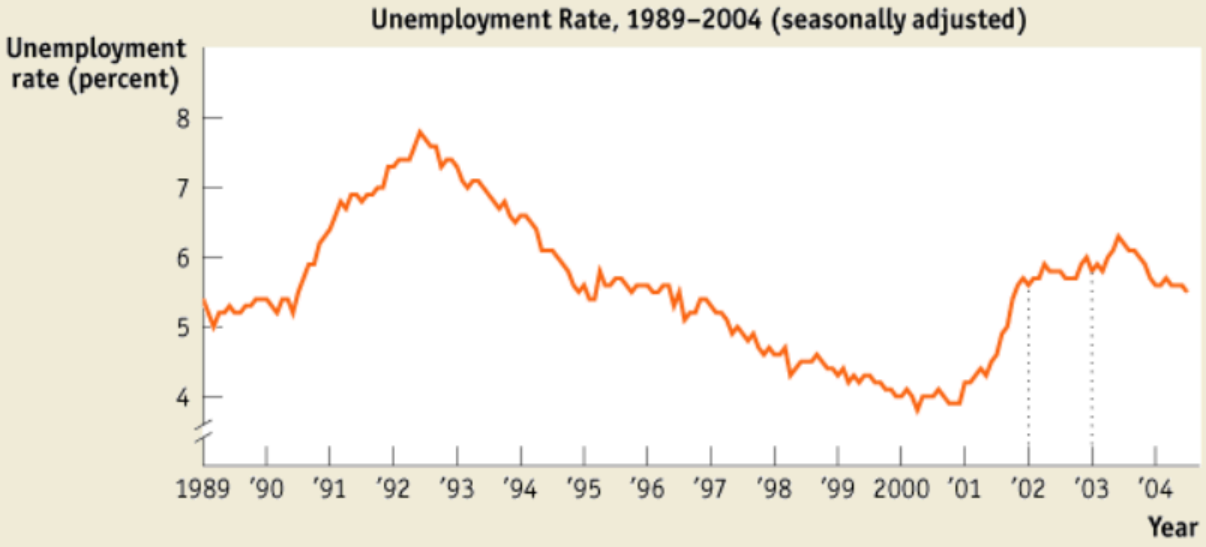
A **time-series graph** has dates on the horizontal axis and values of a variable that occurred on those dates on the vertical axis.

Figure 2A-6

Time-Series Graph

Time-series graphs show successive dates on the *x*-axis and values for a variable on the y-axis. This time-series graph shows the seasonally adjusted unemployment rate in the United States from 1989 to mid-2004.

Source: Bureau of Labor Statistics.



Presenting Numerical Information

Cross Sectional Data and Scatter Diagram

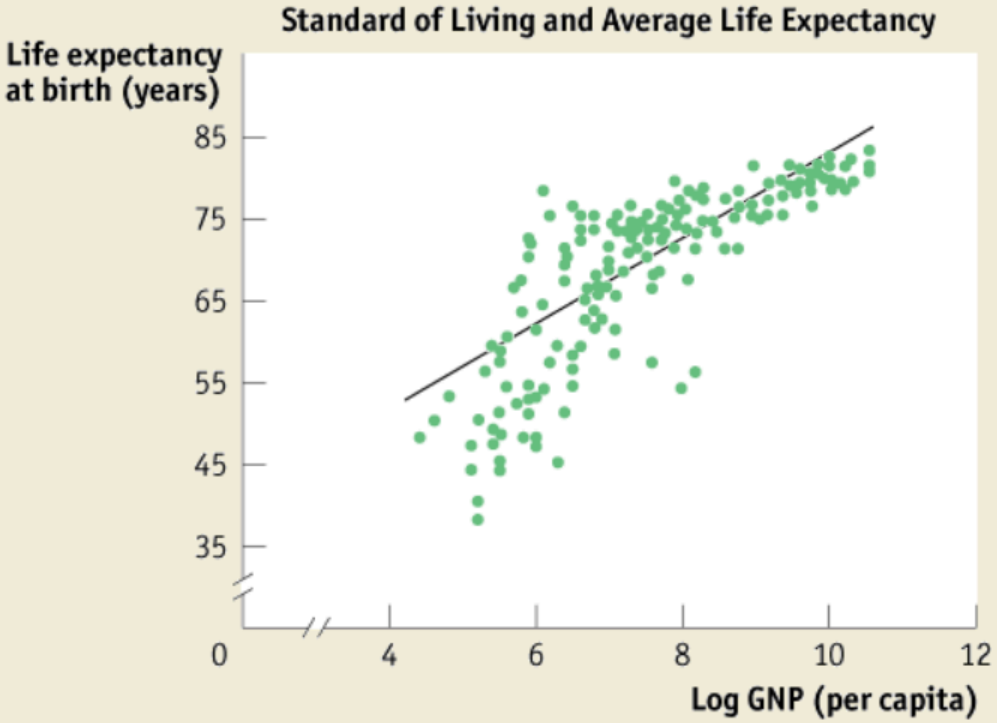
A **scatter diagram** shows points that correspond to actual observations of the *x*- and *y*-variables. A curve is usually fitted to the scatter of points.

Figure 2A-7

Scatter Diagram

In a scatter diagram, each point represents the corresponding values of the *x*- and *y*-variables for a given observation. Here, each point indicates the observed average life expectancy and the log of GNP per capita of a given country for a sample of 158 countries. The upward-sloping fitted line here is the best approximation of the general relationship between the two variables.

Source: Eduard Bos et al., *Health, Nutrition, and Population Indicators: A Statistical Handbook* (Washington, DC: World Bank, 1999).



Presenting Numerical Information

Types of Numerical Graphs

A **pie chart** shows how some total is divided among its components, usually expressed in percentages.

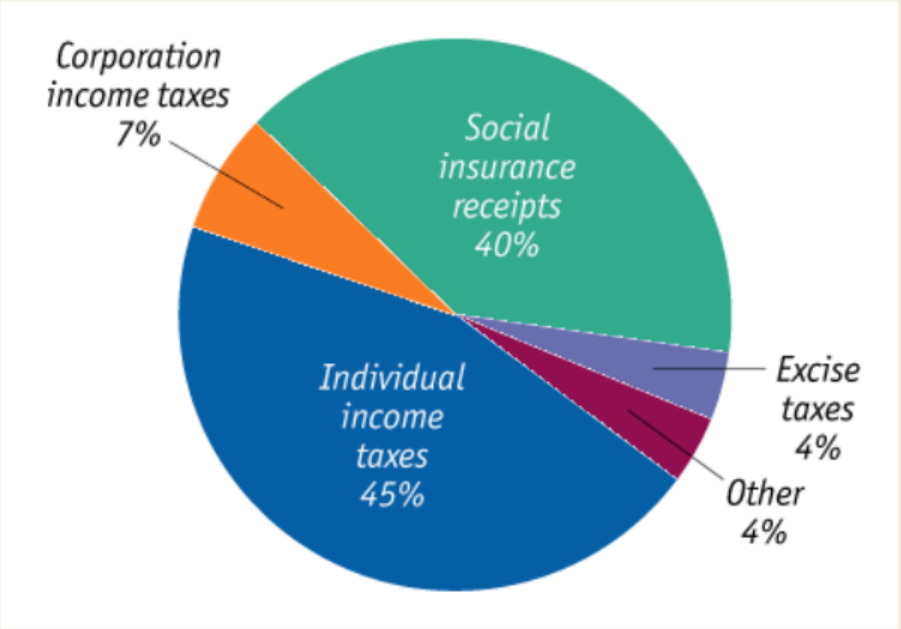
Figure 2A-8

Pie Chart

A pie chart shows the percentages of a total amount that can be attributed to various components. This pie chart shows the percentages of total federal revenues that come from each source.

Source: Executive Office of the President, Office of Management and Budget.

Receipts by Source for U.S. Government Budget 2003 (total: $1,782.3 billion)



Presenting Numerical Information

Types of Numerical Graphs

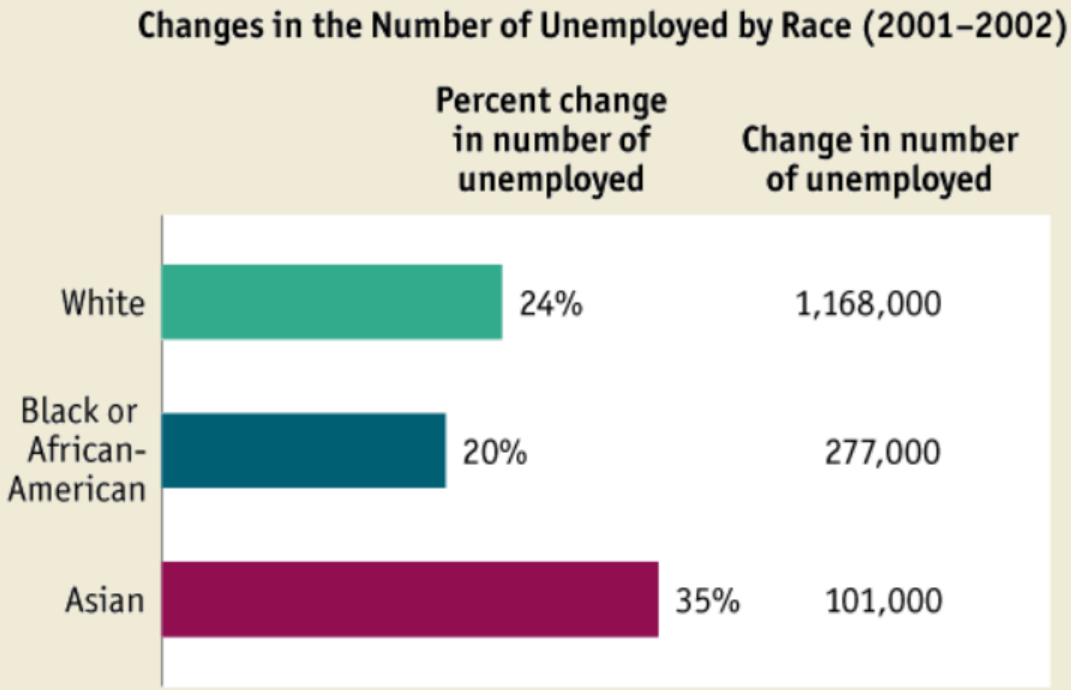
A **bar graph** uses bars of varying height or length to show the comparative sizes of different observations of a variable.

Figure 2A-9

Bar Graph

A bar graph measures a variable by using bars of various heights or lengths. This bar graph shows the percent increase in the number of unemployed between 2001 and 2002, separately for White, Black or African-American, and Asian workers.

Source: Bureau of Labor Statistics.



Presenting Numerical Information

Problems in Interpreting Numerical Graphs

Features of Construction

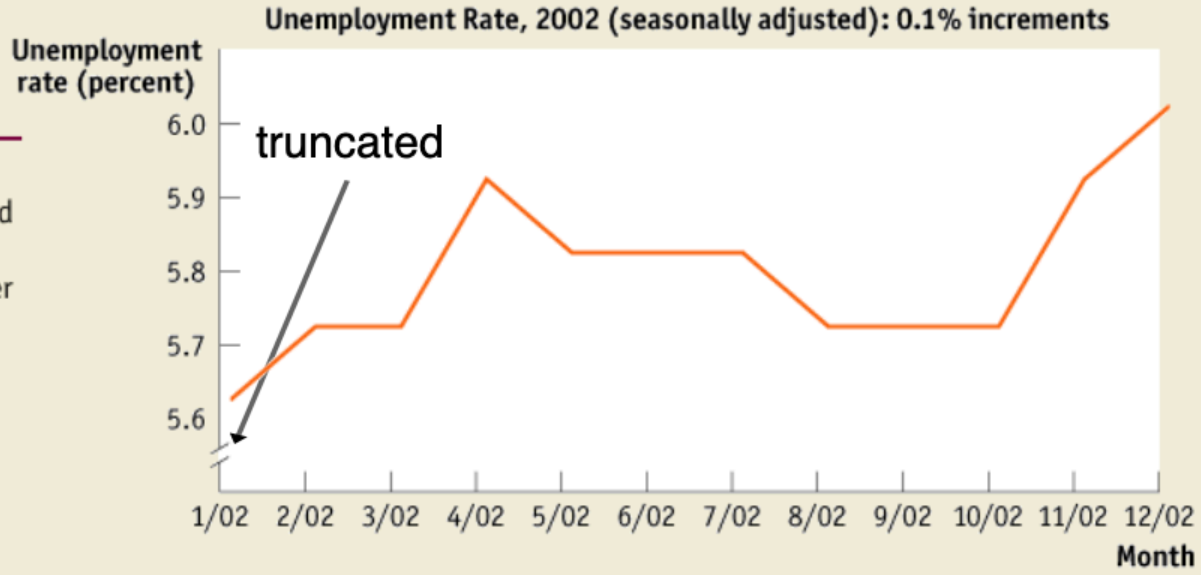
An axis is **truncated** when some of the values on the axis are omitted, usually to save space.

Figure 2A-10

Interpreting Graphs: The Effect of Scale

Some of the same data for 2002 used for Figure 2A-7 are represented here, except that here they are shown using 0.1% increments rather than 1% increments. As a result of this change in scale, the rise in the unemployment rate during 2002 looks much larger in this figure compared to Figure 2A-7.

Source: Bureau of Labor Statistics.



Presenting Numerical Information

Problems in Interpreting Numerical Graphs

Features of Construction

An axis is **truncated** when some of the values on the axis are omitted, usually to save space.

Figure 2A-10

Interpreting Graphs: The Effect of Scale

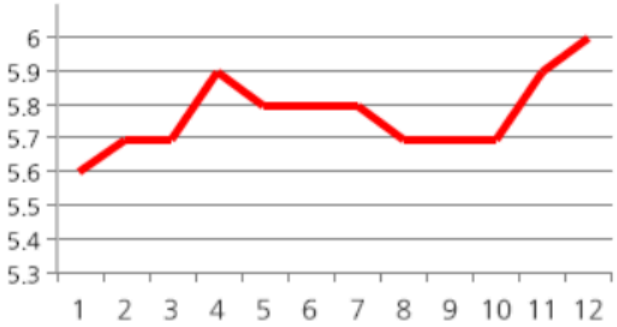
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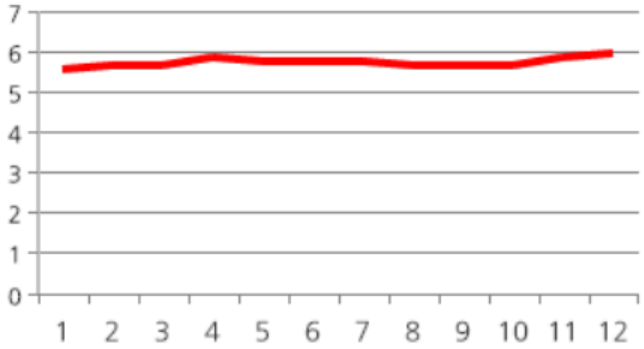
Source: Bureau of Labor Statistics.



The Same Data but Different Looking Graphs







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Dependent Variable and Independent Variable

Dependent Variable: Effect, Resulting Variable
Independent Variable(s): Cause, Causing variables

Ex) Quantity of Money → Inflation
Income → Consumption
Indicates the direction of causation
cause effect

→

Presenting Numerical Information

Problems in Interpreting Numerical Graphs

Omitted Variables

An **omitted variable** is an unobserved variable that, through its influence on other variables, creates the erroneous appearance of a direct causal relationship among those variables.

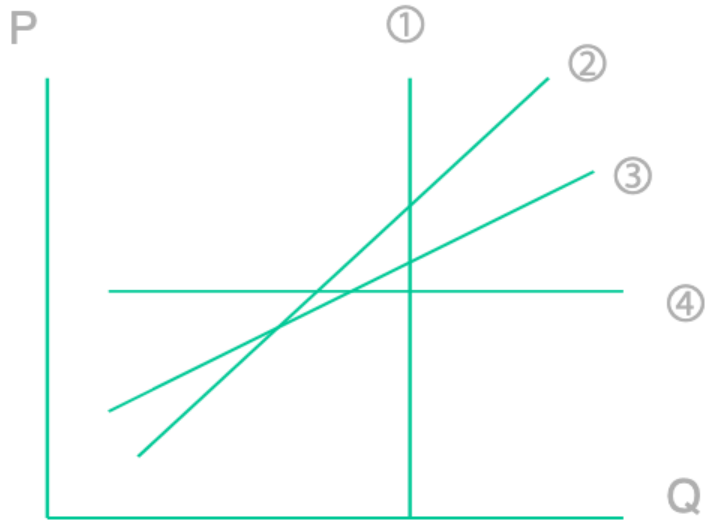
ex) High attendance causes high grade? High attendance may only reflect studying hard.

Reverse Causality

The error of **reverse causality** is committed when the true direction of causality between two variables is reversed.

ex) Income ⇄ Health High income causes good health, but good health can bring in high income.

Elastic and Inelastic



The Change of Q in respond to the change of P

① : No elastic

② : Elastic

③ : More elastic

④ : Infinitely elastic

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